



An efficient piezoelectric single-crystal L-argininium phosphite: structural, Hirshfeld, electrical and mechanical analyses for NLO applications

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Abstract

L-Argininium phosphite, nonlinear optical crystal was grown by the slow evaporation solution growth technique. It crystallizes in the monoclinic system, with $P2_1$ space group and dimensions of unit cells are $a = 9.3677 \text{ \AA}$, $b = 4.8601 \text{ \AA}$ and $c = 13.2096 \text{ \AA}$. Its morphology was simulated and indexed using WinXMorph program. The contribution of different interactions in the crystal structure was discerned visually using Hirshfeld surface analysis. The positions of NMR signal (δ) corresponding to carbon and hydrogen atoms in titled molecule were recorded using HETCOR analysis. Moreover, the titled compound's crystallinity was examined by performing omega scan. Through UV–Vis–NIR transmittance spectra, the cutoff wavelength and bandgap for titled crystal were evaluated. Its dielectric constant and losses were investigated as frequencies varied at room temperature. Furthermore, laser and shock damage threshold analysis were employed on titled compound. Its piezoelectric charge coefficient was also determined and found to be 12 pC/N along (010) plane. Kurtz and Perry's technique was used to measure its second harmonic generation (SHG) efficiency, which was observed to be 20 mV taking standard KDP as reference.

1 Introduction

Today nonlinear optical single crystals are responsible for outstretched applications of the solid-state lasers which extend its wavelengths by cascaded frequency conversion [1]. Various technological areas, viz. information communications, medical treatments and material processing depend on coherent light-producing capability of these lasers [2]. Hence, these

materials play a major role in substituting conventional electronics with their optical analogue such as modulators, amplifiers, rectifiers and switches for optoelectronics and photonics applications [3]. The key requirement that is noncentrosymmetry or absence of inversion symmetry in materials offers unprecedented properties (symmetry dependent) such as pyroelectricity, piezoelectricity, ferroelectricity, and second harmonic generation, which are the source of various applications [4–10]. Most of the amino acids interact nonlinearly with high-intensity laser radiations in terms of polarization and electric field. This behaviour has been observed due to the presence of more than one functional group that introduces chirality within the molecule. The investigations of physical properties of such type of crystals or having multiple-functional groups provide a correlation between its structure and different properties [11]. Till now, a number of organic and inorganic materials such as L-arginine 4-nitrophenolate 4-nitrophenol dihydrate, L-arginine *para*-nitrobenzoate monohydrate, L-arginine acetate, L-argininium bis(trifluoroacetate), L-arginine hydrobromide monohydrate have been synthesized and characterized [12–16]. It has been observed that the organic compounds show high nonlinear efficiency and more sensitivity to electro-optic effect as compared to inorganic

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compounds due to delocalization of π electrons [17–19]. But their practical application was confined due to its fragile nature that creates difficulties in producing single crystal of large size. Besides that, inorganic materials exhibit excellent mechanical properties and high melting point but they show low optical nonlinearities [20]. Aforesaid limitations can be solved by developing a new class of hybrid NLO materials in which high nonlinear efficient organic compounds make a strong ionic bond with an inorganic counterpart [21].

The readily available precursors for the salt synthesis of such class of materials are amino acids and strong inorganic acids. Most of them crystallize in the non-centrosymmetric crystal structure. L-Arginine is an organic amino acid that can be derivatized into semi-organic salts such as L-arginine phosphate, L-arginine hydrobromide monohydrate, L-arginine hydrochloride to produce high nonlinearities and mechanical strength [22, 23]. L-Argininium phosphite (LAPI) falls into the same category which was first reported by Ghazaryan et al. in [26], which crystallizes in non-centrosymmetric crystal system having $P2_1$ space group [24]. However, to date, there has been no report on piezoelectricity and nonlinear efficiency of the titled compound, and thus the focus of the work is to explore these properties of the material. On experimental investigations, a sufficiently good piezoelectric response was obtained and was also found to be NLO active. Thus, LAPI crystal is an appropriate material for image processing, ultrasonic transducers, ignitors, actuators, impact detector and position sensor applications [25]. The contributions of different interactions in the packing of molecular crystal were determined using Hirshfeld surface analysis [26]. HETCOR was carried out to find out positions of NMR signal (δ) corresponding to carbon and hydrogen atoms in titled molecule. Mechanical strength of (001) plane of titled crystal was evaluated using Vickers microhardness technique and Meyer index (n) was calculated to understand its nature. Thus, the current research article reports various important studies such as structural, physical, optical, crystalline perfection, mechanical, piezoelectric and electrical properties of a NLO crystal and proves it to be a potential material for various device fabrications.

2 Experimental section

2.1 Salt synthesis and growth of single crystal

L-Arginine ($C_6H_{14}N_4O_2$) and phosphorous acid (H_3PO_3) (purchased from Sigma-Aldrich) were mixed in equimolar ratio in a beaker containing distilled water until saturation [24]. The resulting solution was stirred at 30 °C for 5 h to obtain homogeneous and transparent solution. The schematic view of the above reaction is shown in Scheme 1. The resulting transparent solution was kept in a beaker covered with perforated aluminium foil for slow evaporation at 30 (± 0.1 °C) in a constant temperature bath. Initially, needle-shaped poor-quality crystals were obtained due to the high viscosity of the solution. After repeated recrystallization of about seven to eight times, single crystals embedded inside thick polycrystalline layer were harvested. To extract these crystals, above layers were carefully etched out. The plate-shaped colourless crystals were obtained from the prepared saturated solution after a time period of 16 days [24]. The photograph of as-grown crystal is presented in Fig. 1a, b. Figure 1c shows the morphology of as-grown LAPI crystal, which was simulated and indexed using WinXmorph program [27].

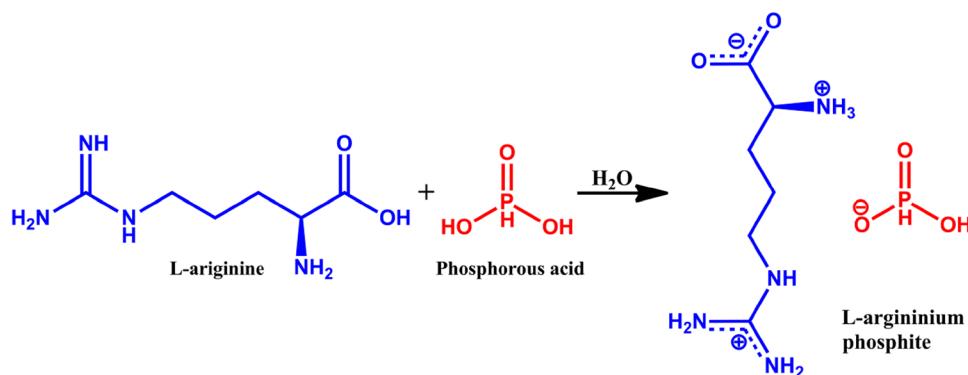
Elemental (CHNS) analysis was performed on LAPI crystal ($C_6H_{17}N_4PO_5$; M_w : 256.21) using CHNS analyzer, in which the experimental results (C = 28.17%, H = 6.61%, and N = 21.92%) were found to be matching with the calculated values (C = 28.13%, H = 6.69% and N = 21.87%).

3 Characterizations

3.1 Single-crystal XRD and powder XRD

The crystallographic parameters of as-grown LAPI crystal were studied by SCXRD analysis. Good quality and twin-free crystals of size $0.2 \times 0.2 \times 0.2$ mm³ were chosen to collect the structural information. The molecular structure of

Scheme 1 Synthesis of L-argininium phosphite salt



PXRD data were collected using finely crushed powder of single crystal and subjected to Rigaku Ultima IV multiple

In ^1H NMR of the compound, the singlet at δ 7.33 ppm for 0.5 H and 5.77 ppm for 0.5 H corresponds to H-1'. A triplet at δ 3.49 ppm with coupling constant (J) 6.1 Hz for 1H corresponds to the protons of C-2 due to splitting with

Table 1 Crystal data and structure refinement for the LAPI single crystal

Empirical formula	C ₆ H ₁₇ N ₄ O ₅ P
Formula weight	256.21
Temperature/K	293 (2)
Crystal system	Monoclinic
Space group	P2 ₁
<i>a</i> /Å	9.3697 (10)
<i>b</i> /Å	4.8601 (6)
<i>c</i> /Å	13.2096 (14)
<i>α</i> /°	90.00
<i>β</i> /°	105.418 (11)
<i>γ</i> /°	90.00
Volume/Å ³	579.89 (11)
<i>Z</i>	2
ρ_{calc} g/cm ³	1.467
μ /mm	0.252
<i>F</i> (000)	272.0
Crystal size/mm ³	0.2×0.2×0.2
Radiation	MoK α (λ =0.71073)
2 θ range for data collection/°	8.74–52
Index ranges	−11 ≤ <i>h</i> ≤ 8, −5 ≤ <i>k</i> ≤ 5, −14 ≤ <i>l</i> ≤ 16
Reflections collected	2324
Independent reflections	1891 [<i>R</i> _{int} =0.0219]
Data/restraints/parameters	1891/1/213
Goodness-of-fit on <i>F</i> ²	1.052
Final <i>R</i> indexes [<i>I</i> ≥ 2 σ (<i>I</i>)]	<i>R</i> ₁ =0.0365, <i>wR</i> ₂ =0.0926
Final <i>R</i> indexes [all data]	<i>R</i> ₁ =0.0386, <i>wR</i> ₂ =0.0946
Largest diff. peak/hole/e Å ^{−3}	0.23/−0.36
Flack parameter	0.79 (13)

neighbouring protons attached to the C-3 carbon. Similarly, a triplet at δ 2.95 ppm with coupling constant (*J*) 6.9 Hz is for 2H of C-5 due to splitting with neighbouring protons attached to C-4. A multiplet at δ 1.65–1.60 ppm for 2H of C-3

carbon arises because of splitting with neighbouring protons attached to C-2 and C-4 carbon. In the same way, a multiplet at δ 1.47–1.31 ppm is for 2H of C-4 carbon which is due to splitting with neighbouring protons attached to C-3 and C-5. In ¹³C NMR of the compound, the peak at δ 174.40 ppm corresponds to C-1 and peak at δ 156.75 ppm corresponds to C-7. The peaks at δ 54.25, 40.46, 27.52 and 23.86 ppm are corresponding to carbons C-2, C-5, C-4, and C-3, respectively.

3.3 Hirshfeld surface analysis

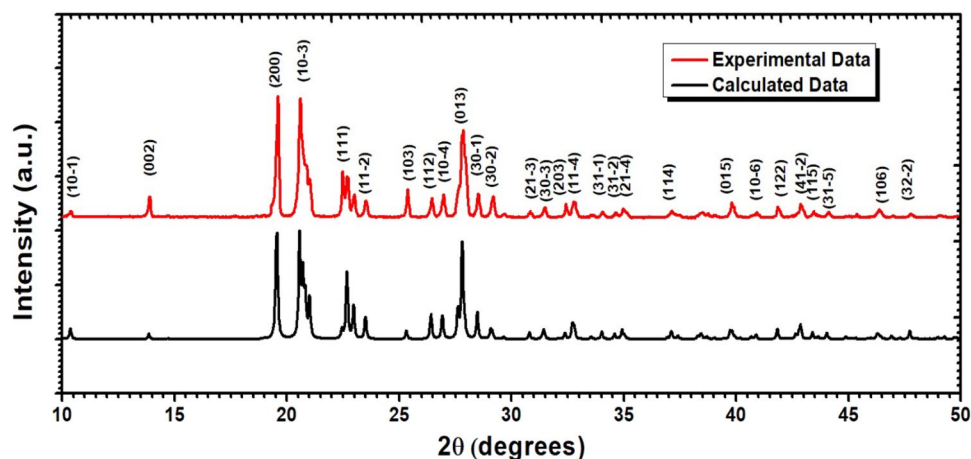
The pattern of hydrogen bonding and metal coordination helps to decide the solid-state behaviour of molecules through supramolecular chemistry [31]. The relative strength of the hydrogen bond is weaker than other interactions. Thus, the supramolecular packing design of the molecules is easily altered by the various possible combinations of hydrogen bonds in the crystal lattice. The involvement of various atomic interactions in the material structure is discerned visually using Hirshfeld surface analysis [32]. A whole-of-molecule approach is required to generate the Hirshfeld surface using 0.5 isosurfaces of weight function *w* (*r*) of the promolecule density [33]. The weight function for Hirshfeld surface is defined as the ratio of the sum of spherical atom electron densities for a molecule (promolecule) of interest and the same sum for the crystal (procrystal):

$$w(r) = \frac{\sum_{i \in \text{molecule A}} \rho_i(r)}{\sum_{i \in \text{crystal}} \rho_i(r)} \quad (1)$$

$$= \rho_{\text{promolecule}}(r) / \rho_{\text{procrystal}}(r).$$

To highlight the close contacts between larger atoms, normalized contact distance d_{norm} is used to generate the Hirshfeld surface:

$$d_{\text{norm}} = \frac{d_i - r_i^{\text{vdW}}}{r_i^{\text{vdW}}} + \frac{d_e - r_e^{\text{vdW}}}{r_e^{\text{vdW}}}, \quad (2)$$

Fig. 3 Experimental and calculated powder X-ray diffraction pattern

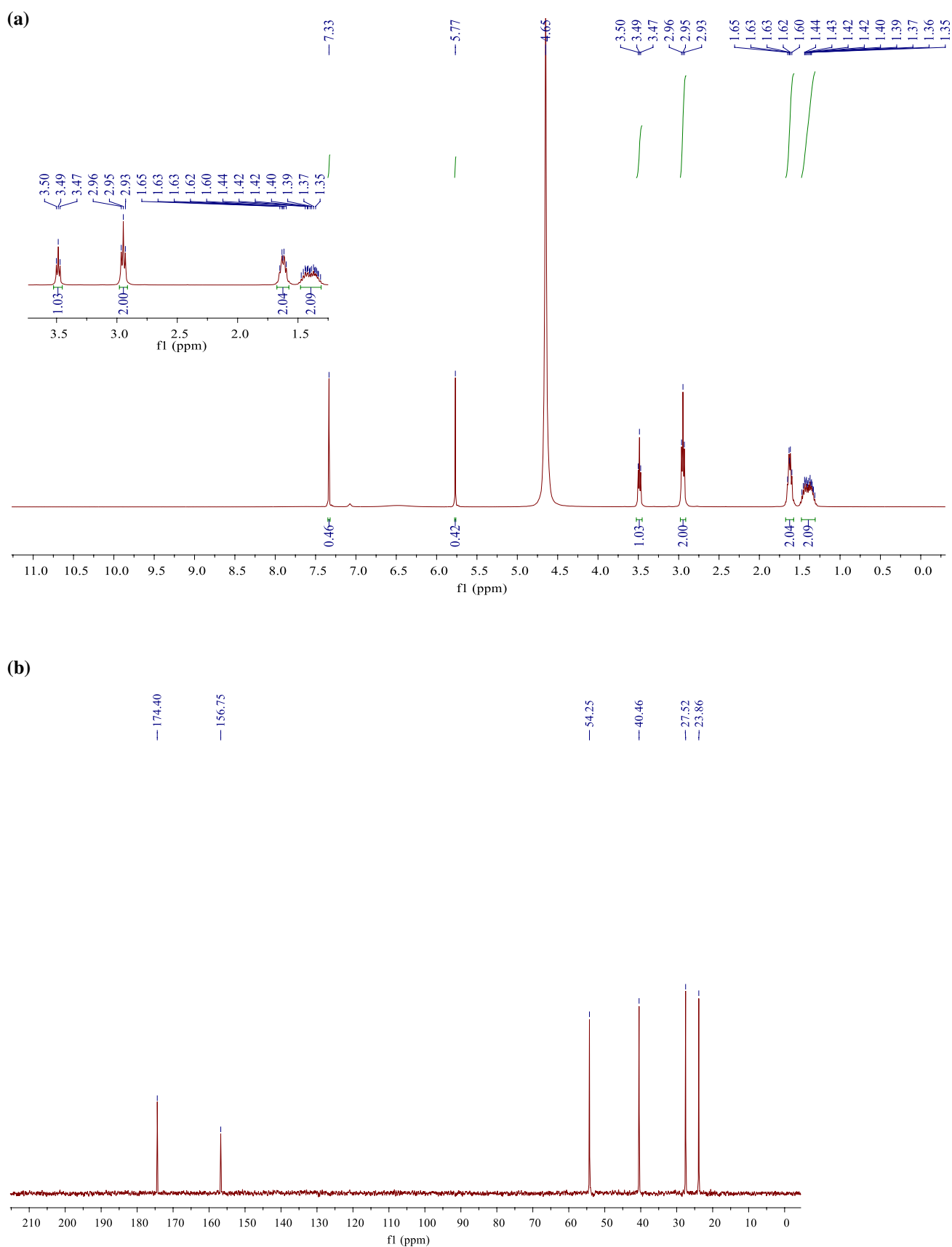


Fig. 4 **a** ^1H NMR spectrum of the compound LAPI (400 MHz, D_2O). **b** ^{13}C NMR spectrum of the compound LAPI (100 MHz, D_2O)

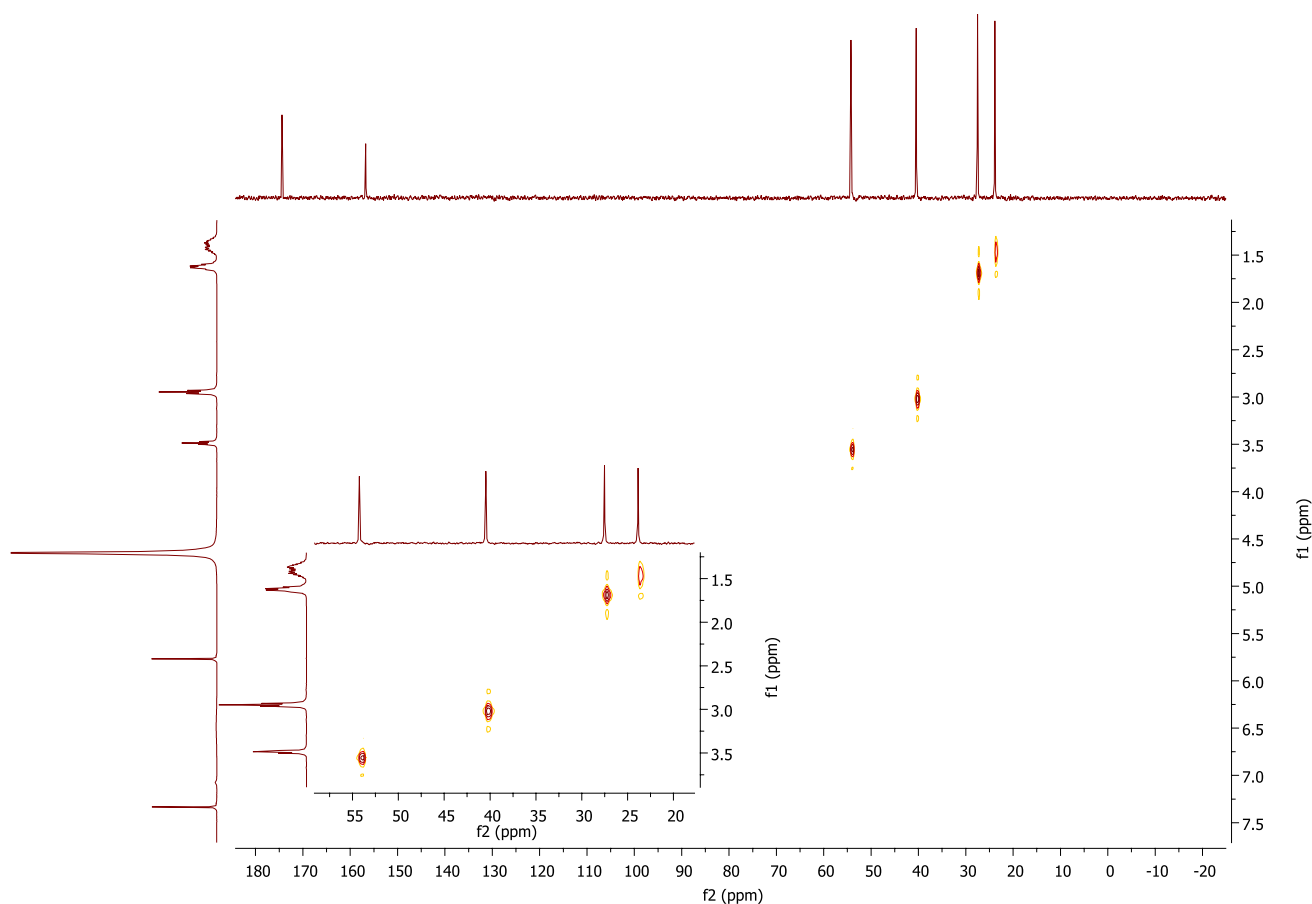


Fig. 5 HETCOR of the compound LAPI

where r^{vdW} is the van der Waals (vdW) radius of the atom, d_e represents the distance to the nearest atoms from the interior of the molecular surface and d_i shows the distance to the nearest atoms from the exterior of the molecular surface [26]. In addition, d_{norm} is also symmetric in d_e and d_i .

Figure 6a, b illustrates the front and back views of molecular Hirshfeld surface mapped over d_{norm} for the L-argininium cation of the grown crystal along with the neighbouring molecules. The red colour regions on the Hirshfeld surface represent the strong O $\cdots$ H contacts, which relate to various physical properties of the crystalline material with anisotropic nature. The fingerprint plots quantify the contribution of all intermolecular interactions present in the crystal lattice. The major interactions in the L-argininium cation come from the strong O $\cdots$ H (47.7%) and weak H $\cdots$ H (41.5%) contacts, respectively. The O $\cdots$ H interactions play a noteworthy role in the piezoelectric nature of the LAPI crystal. The N $\cdots$ H, C $\cdots$ H, N $\cdots$ O and O $\cdots$ C intermolecular interactions in L-argininium cation of titled crystal are displayed in Fig. 6c–f with 5.0%, 4.0%, 1.0% and 0.5% areas of fingerprint plot, respectively.

Figure 7a, b represents the front and back views of molecular Hirshfeld surface mapped over d_{norm} for the phosphite anion of the titled crystal along with the neighbouring molecules. In the case of fingerprint plot, the two sharp peaks correspond to the O $\cdots$ H interactions with the red colour region on the Hirshfeld surface of phosphite anion in LAPI crystal. The H $\cdots$ H and N $\cdots$ O interactions in phosphite anion are shown in Fig. 7c–e with 25.6%, and 1.6% areas of the fingerprint plot, respectively.

3.4 Crystalline perfection analysis

Crystal quality or perfection of the specimen was assessed using a PANalyticalX'Pert PRO MRD high-resolution XRD system, having $\text{CuK}_{\alpha 1}$ as a radiation source. The rocking curves (RC) for different diffracting planes of the specimen were measured by varying the angle between the crystal surface and incident beam (during ω scan, the detector position was fixed at $2\theta_B$, i.e. 19.4°) with respect to Bragg diffraction peak position at an angle of 9.7° . The monochromated X-ray beam was achieved using high-resolution four-bounce

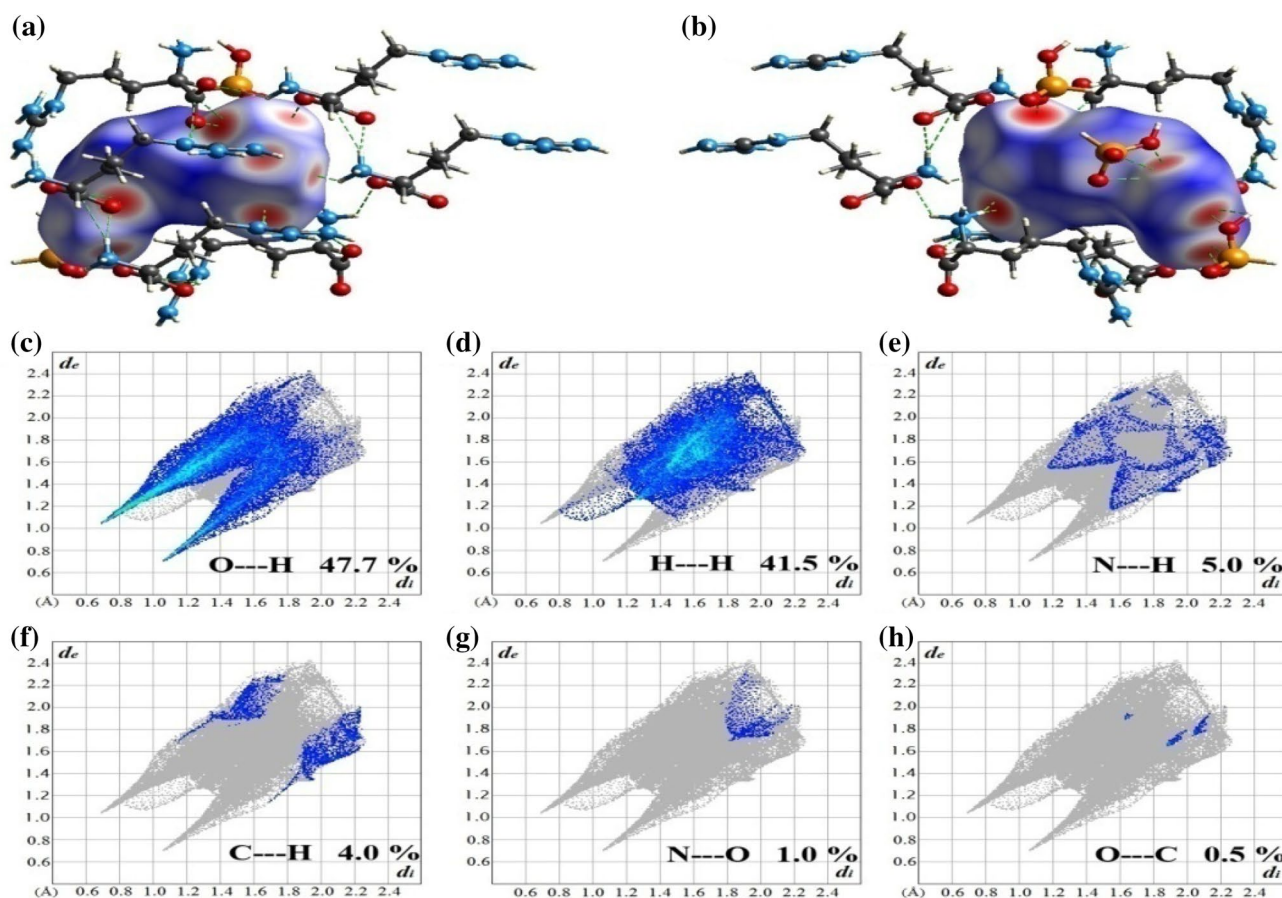


Fig. 6 **a, b** Front and rear views of Hirshfeld surface mapped with d_{norm} for the L-argininium cation. **c–h** Fingerprint plots depicting all the contribution of each intermolecular interactions of L-argininium cation with neighbouring molecules

monochromator, i.e. Ge (220). The obtained beam was further made to fall on the crystal. A scintillator detector was used to detect diffracted beam [34].

Before starting the measurement, first, the crystal was lapped to make its surface flat and to exterminate any impurities present on the crystal surface using a diluted solution of acetone [28]. Figure 8 represents the high-resolution diffracting curve recorded along diffracting plane (200) in symmetrical Bragg geometry of LAPI crystal.

The curve encloses a single sharp peak with a full width at half maxima (FWHM) of 46 arc seconds. That means the crystals neither have low- and high-angle structural grain boundaries nor the formation of the epitaxial layer which is usually found in solution-grown crystals [35]. Though the peak obtained is sharp, the observed FWHM is higher than that expected from plane wave theory for model crystals [36, 37]. This suggests the slight deviation from ideal crystals and thus minor defect may exist having low defect concentration. Such crystals of low defect concentration are adaptable for practical application because no sufficient degradation of properties is caused by these defects [38].

If we see the curve carefully, it has been observed that the scattered intensity in positive direction is more as compared with the opposite direction signifying that crystal predominantly comprises interstitial defects. This is due to the fact that lattice around interstitial type defect undergoes compression leading to higher scattered intensity in a positive direction with respect to zero peak position due to decrease in inter-planar spacing in accordance with Bragg's law. The converse explanation is true for vacancy type of defect where lattice around point defect undergoes expansion resulting in an increase in inter-planar spacing " d ", hence, resulting in higher scattered intensity in a negative direction with respect to Bragg's peak position [39]. Overall, the crystalline perfection of LAPI crystal is fairly good as evident from obtained FWHM; hence, it can be utilized for NLO applications.

3.5 Transmission studies

Transmission range in the UV–Vis–NIR region not only decides the material application in optoelectronic devices, it also helps in structural elucidation of a material. When a

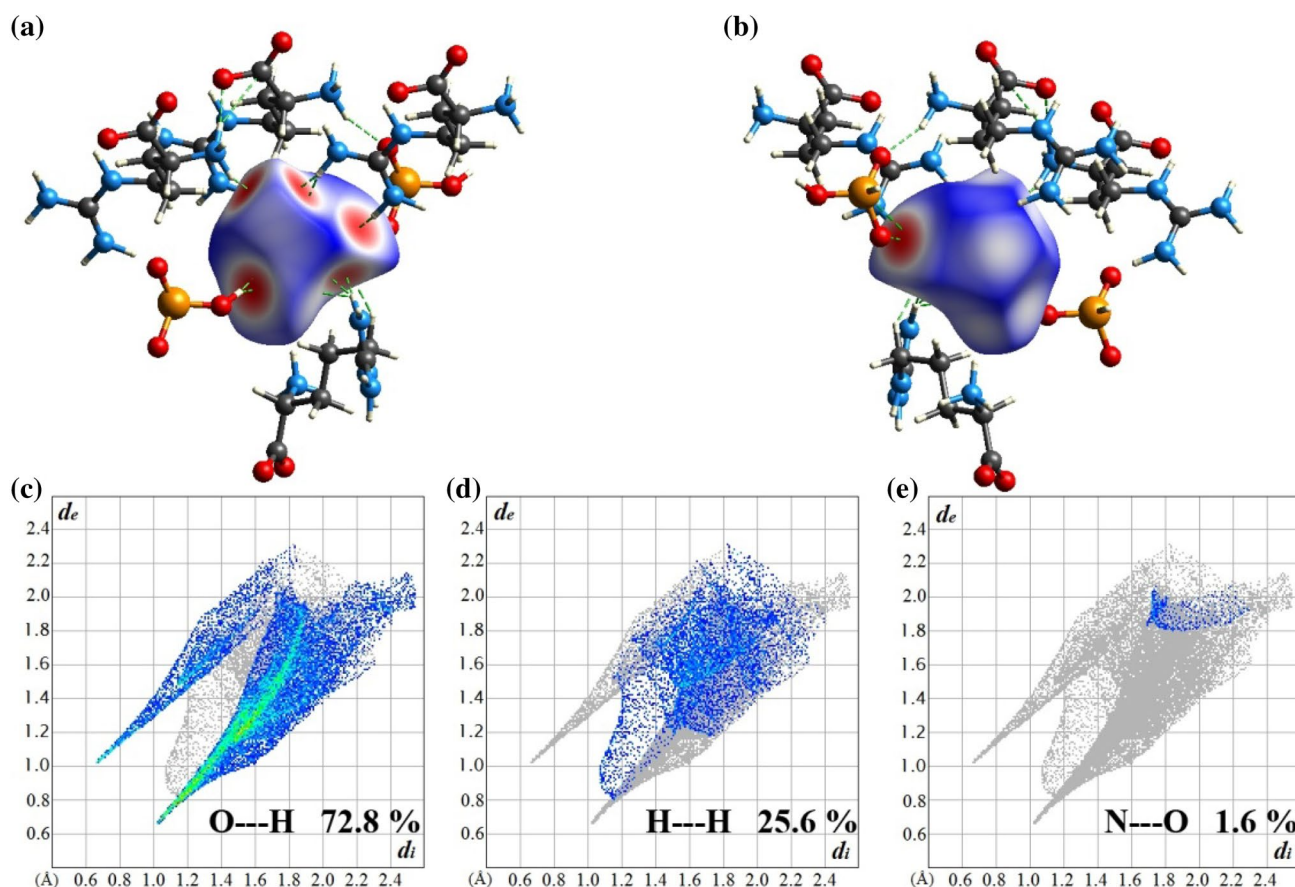


Fig. 7 **a, b** Front and rear views of Hirshfeld surface mapped with d_{norm} for the phosphite anion. **c–e** Fingerprint plots depicting all the contributions of each intermolecular interaction of phosphite anion with neighbouring molecules

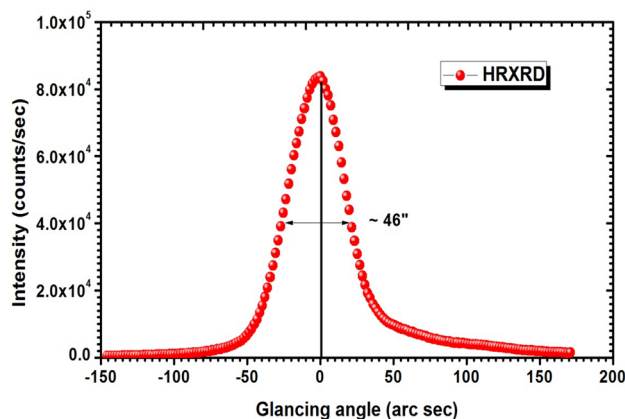


Fig. 8 Rocking curve recorded for LAPI crystal

material shows electron transitions of π and σ orbitals due to absorption of UV–Vis–NIR light, various structural information can be obtained [40]. The optical transmission data of L-argininium phosphite single crystal of 1 mm thickness shown in Fig. 9a have been recorded in the wavelength range of 200–1100 nm.

From this curve, one can determine the lower UV cutoff wavelength and for the titled material, it was found to be 240 nm approximately. The optoelectronic efficiency of NLO materials depends on its transparency percentage in the region of UV–Vis–NIR. The titled compound can be utilized for SHG applications as it exhibits good transmittance about 62.5% at 532 nm. The various optical parameters play a significant role in its implementation in an optical device [41, 42]. The type of optical transitions in any material is governed by the relationship between optical absorption coefficient and photon energy. The optical absorption coefficient can be calculated from the transmission data using the equation

$$\alpha = \frac{2.303 \times \log \frac{1}{T}}{d}, \quad (3)$$

where d and T have their usual meanings. Using Tauc relation one can estimate the optical bandgap (E_g) of the material from the optical absorption coefficient (α) near the absorption edge [43]:

$$\alpha h\nu = \frac{A(h\nu - E_g)^{\frac{1}{2}}}{h\nu}. \quad (4)$$

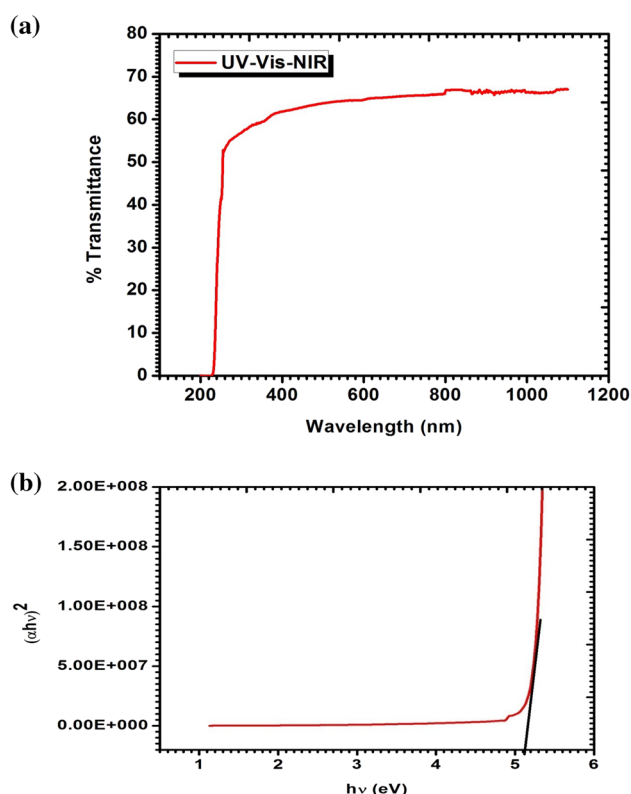


Fig. 9 **a** Transmission spectrum of LAPI crystal. **b** Plots of $(ah\nu)^2$ function of $h\nu$ with the evaluation of bandgap of the material

Further, a graph has been plotted between $(ah\nu)^2$ and photon energy ($h\nu$) and a tangent was drawn that cuts the X -axis, i.e. photon energy axis ($h\nu$), which depicts the bandgap of material (see Fig. 9b).

The optical bandgap is found to be nearly 5.1 eV.

3.6 LDT measurements

Nd:YAG nanosecond laser was used to quantify the crystal quality at a wavelength of 1064 nm. The width of above laser beam pulse was set at 10 ns which was Gaussian in shape (3.5 mm diameter) and the repetition rate was set at 1–10 pulses per second, respectively. A bi-convex lens (20 cm focal length) was used to focus the exposed laser beam. The laser was focused using a bi-convex lens of focal length 20 cm. In this study, a well-polished crystal wafer was positioned in the beam pathway at the focal point of the lens. This technique involved a subsequent increase in applied voltage of incident laser and resistance of crystal against beam exposure was

measured until the crack was seen at the crystal surface. Using the value of input laser source energy (E) and area of the circular spot (A), the energy density was calculated [44, 45]. The crystal was subjected to two different positions to record its damage threshold at the single shot and multiple shots of laser beam exposure. It has been noticed that whenever a material was irradiated with laser exposure, it generates heat within the material that results in damage to the crystal. Hence, a material can show resistance against laser beam for the longer time period with high specific heat (smaller temperature gradient) and vice versa [46]. Moreover, its laser tolerance capacity also depends on crystalline perfection or crystal quality of the sample. The defect concentration will reduce its mechanical strength because these defects act as energy-absorbing centres [47]. In the present study, the observed value of damage threshold is tabulated in Table 2.

From the table, one can observe that the damage threshold for 10 pulses per second was reduced more due to decline in the mechanical strength of the sample as a result of increasing laser power. This provides an idea about how much laser power should be applied when the crystal is implemented in a particular device application without undergoing any damage or cracks. The LDT value of the titled compound was found to be high when compared to KDP, LAHCL and less than urea, PPN, L-arginine phosphite [45, 48–51]. Hence, grown single crystal exhibits reasonably good LDT value.

3.7 Dielectric studies

Dielectric constant is an essential parameter governing the mechanism of polarization of atoms and ions present in a single crystal. Measurement of dielectric properties provides quality information of materials [52]. Dielectric studies on L-argininium phosphite single crystal were measured over a range of frequencies. For that purpose, a good quality transparent crystal has been chosen. The opposite sides of the sample were layered with silver paste to make electrical contact and leave it at 50 °C on hot plate to dry. Dielectric behaviour of any material is associated with its polarization. Moreover, this polarization is found to be dependent on applied field and nature of the various components, i.e. atoms, ions, etc. present within the solids. The dielectric constant can be found out by the following relation:

$$\epsilon' = \frac{Cd}{\epsilon_0 A}, \quad (5)$$

Table 2 Laser damage values of LAPI single crystal for single shot and multiple shots

Crystal	Damage threshold			
	At 1 pulse per second		At 10 pulses per second	
L-Argininium phosphite	6.3 J/cm ²	0.63 GW/cm ²	5.5 J/cm ²	0.55 GW/cm ²

where d is the thickness and A is area of the title compound. Figure 10a, b shows the trend of dielectric behaviour with respect to different frequency ranges at room temperature.

From the curve, one can clearly observe that the dielectric constant value is higher at low frequency which may be due to the contribution of four types of polarizations, namely space charge, orientational, ionic and electronic polarizations. Dielectric constant gradually decreases from 58 to 22 as frequency increases. It saturates at higher frequencies which may be due to diminished contribution of polarization (mainly space charge) from different constituents [53]. The trend of dielectric loss with various frequencies is shown in Fig. 10b. At high frequency, the loss is found to be of low value depicting the better uniformity in crystal structure which is indispensable for laser applications [54].

3.8 Shock damage threshold (SDT)

Shock waves are nonlinear destructive waves that are produced as a result of abrupt discharge of energy in any form of pressure and temperature and travel with supersonic speeds [55]. Such type of waves can also be generated by falling or striking an object on the crystal. To know the capability of the crystal against these events, SDT has been carried out for the titled crystal. Thus, the sample was exposed to shock waves (with known strength) generated using shock tubes that induce plastic deformation, elastic compression, cracks within the crystals. In the present work, the minimum strength of the shock waves to damage the crystal surface is analysed [56–58]. To perform this experiment, an indigenously developed hand-driven shock tube is used. Figure 11a shows the microscopic (10 ×) image of the crystal before the exposure of shock waves.

It clearly shows that the etch pits are well defined and uniform, which shows that the grown crystal is defect free. In the present experiment, the indigenously developed semi-automatic shock tube can generate shock waves of 1 Mach–5

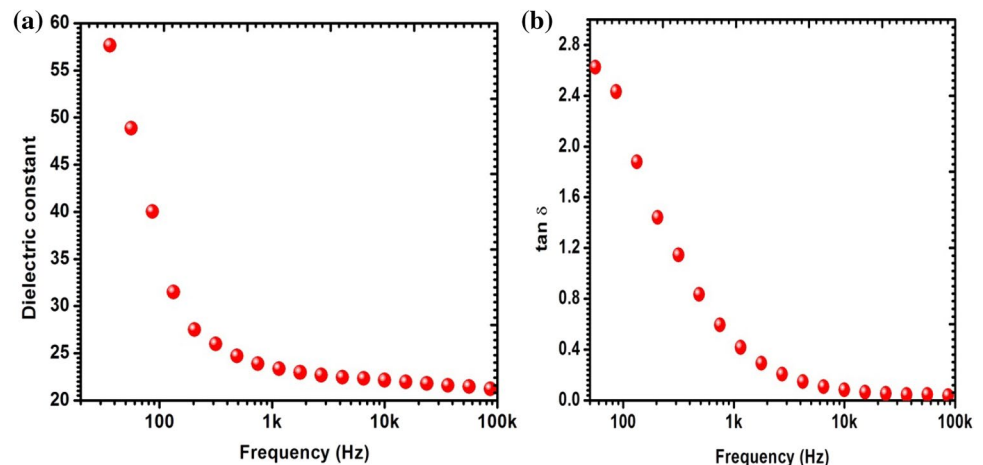
Mach number. The polished and solvent-etched crystal is placed at a distance of 1 cm from the open end of the driven section of the shock tube and shock pulses are loaded on the crystal. The crystal surface is carefully checked using an optical microscope after each shock to confirm the damage. In the present case, a shock of Mach number 2.2 is used for shock damage threshold study. Figure 11b shows the microscopic image of the surface of crystal after second shock. In this case, after the 2nd shock, few pits which resemble the etch pits are observed on the surface of the crystal. When the number of the shock pulses increased, the number and the size of the pits also increased. Interestingly, no cracks are observed even after the 12th shock.

3.9 Mechanical strength

Hardness of a single crystal is evaluated by the resistance of a material to permanent indentation. The study of mechanical strength of solids is valuable for device fabrication and its robust applications. In the case of device fabrication, mechanical strength of the materials is most often considered in the design and mechanical modelling. The Vickers hardness value (H_V) was determined using $H_V = K P/d^2$, where d , P and K stand for average diagonal length of the impression made by indenter, applied load and geometrical constant (1.8544), respectively [59]. The variation of micro-hardness values of the grown crystal with indenter load is displayed in Fig. 12a.

The value of hardness number increases with applied load up to 10 g, which represents the reverse indentation size effect (ISE). It was observed because at initially applied indenter loads, only the softer layers of the crystal surface were pierced by the diamond indenter. Afterwards, a sudden fall of the hardness number has occurred in the region of 10–30 g, which shows the normal ISE behaviour. This observation arises due to the pile-up effect and cracks produced on the crystal surface, which is related to stress released in

Fig. 10 **a** Variation of dielectric constant with frequency at room temperature; **b** variation of dielectric loss with frequency



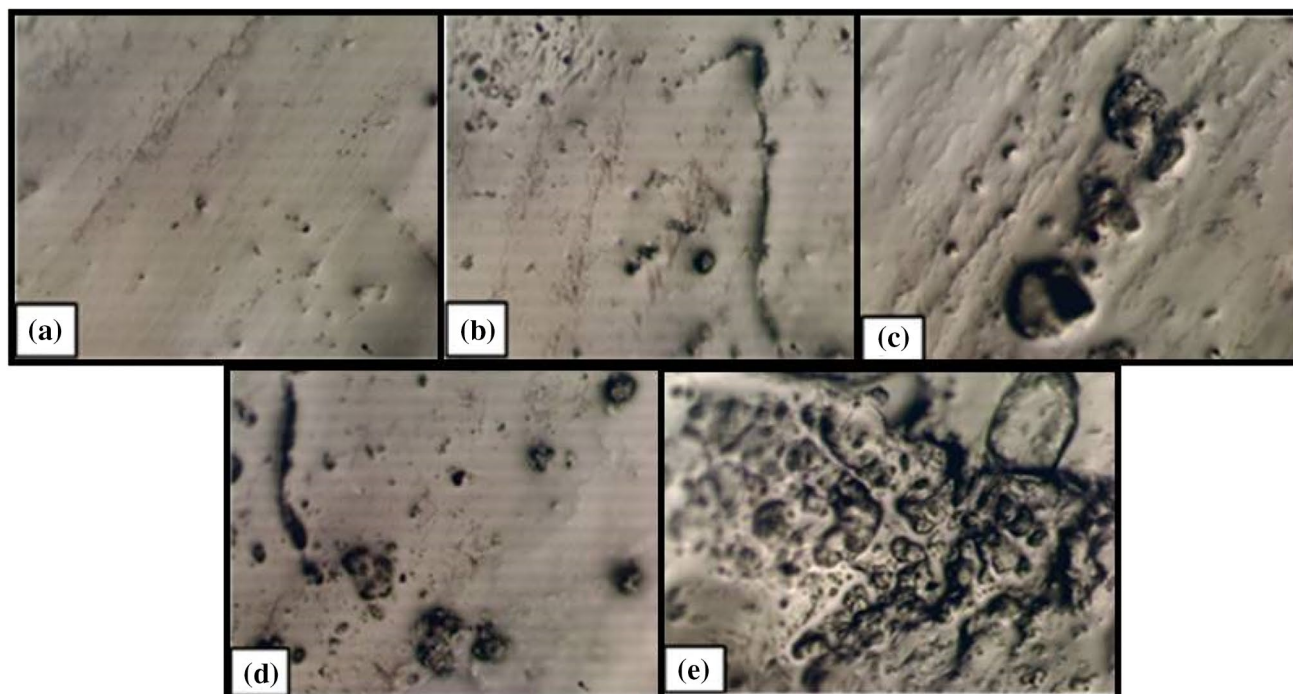


Fig. 11 Optical microscopy images of the crystal surface **a** before applied shocks; **b** after 2nd shock; **c** after 5th shock; **d** after 7th shock; and **e** after 12th shock

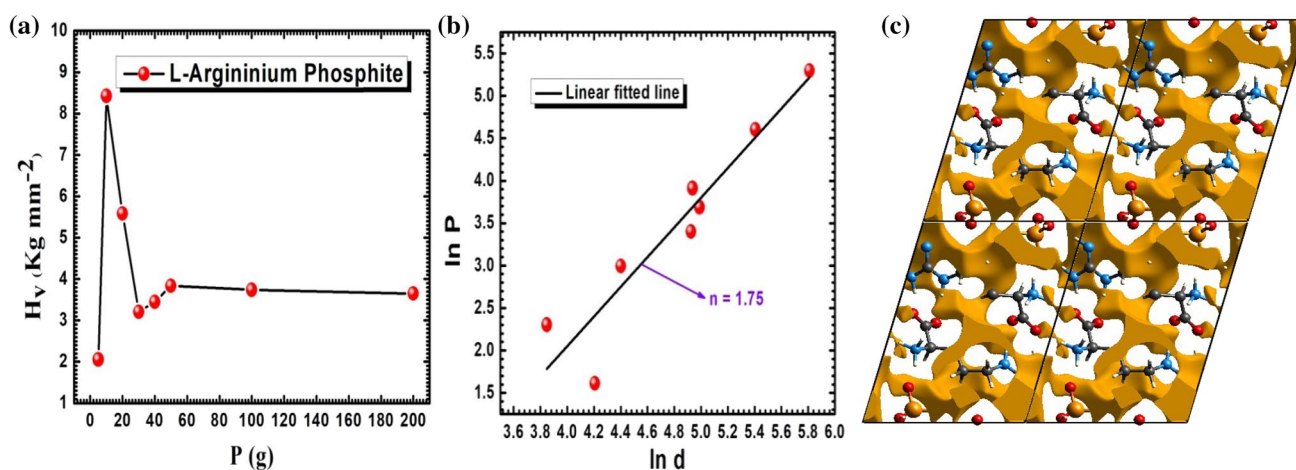


Fig. 12 **a** Plot of hardness number (H_V) versus applied load (P) for the LAPI crystal along (100); **b** the variation of $\ln(P)$ against $\ln(d)$; **c** LAPI unit-cell voids mapped at a 0.002 a.u. isosurface along crystallographic b -axis

the form of cracks. Further, no more variation was observed in the hardness number with applied indenter load, which is due to the mutual interactions of the dislocations.

Meyer's law explains the behaviour of indenter load (P) and diagonal length (d): $P = K d^n$, where K is the work hardening constant and n is Meyer's index number. Plot of $\ln(P)$ versus $\ln(d)$ is shown in Fig. 12b. The value of Meyer's index is evaluated to be 1.75 for LAPI crystal. The observed

value of n is found to be more than 1.6. Thus, as per Ontich's theory, the LAPI crystal belongs to the softer category [60].

Both void space and filled space are equally responsible for the anisotropic characteristics of a crystalline material. The void space in the titled crystal is drawn by rolling a probe sphere of changeable radius over the Hirshfeld surface [60]. Figure 12c shows the many void channels present in the LAPI crystal, which helps in understanding the metal

complex ion doping in host LAPI crystal to enhance its optical and mechanical properties. The 11.60% volume of the unit cell of titled crystal is occupied by voids with surface area of 242.66 Å².

3.10 Piezoelectricity

Piezoelectric crystal produces an electric charge under the action of applied mechanical force. The strength of hydrogen bond along specific directions decides the net dipole moment in the crystal system. Thus, piezoelectric property along particular crystallographic direction leads in character. To measure the piezo-coefficient of the LAPI crystal, silver coating was applied to opposite surfaces to make the electrical contacts and leave it at 40° on hot plate to dry it. The value of piezoelectric coefficient (d_{22}) was found to be 12 pC/N along (010) plane. Its piezoelectric coefficient is found to be higher than quartz (2.17 pC/N) [61], ammonium dihydrogen phosphate (2.92 pC/N) [62, 63] and lithium niobate (10.6 pC/N) [64]. As compared with the other crystals, LAPI crystal has shown promising piezoelectric property which makes it a potential futuristic crystal for sensor, position detector and energy-harvesting applications.

3.11 SHG measurement

The NLO efficiency of the titled compound was scrutinized using Kurtz and Perry powder technique. The asymmetric structure of this compound arises due to the presence of chiral centre which in turn is prerequisite to demonstrate NLO activity. The NLO efficiency depends on the extent of charge delocalisation within material. Organic molecules that contain π orbitals may exhibit such delocalisation. But effective delocalisation can occur if strong intermolecular interactions are present which leads to overlapping of the molecular wave function. Since pure organic crystal has weak interactions, mixing it with inorganic material will introduce strong interaction and results in high optical non-linearity [65, 66]. Nd:YAG laser beam having wavelength 1064 nm and energy 1.0 mJ/pulse was incident on the fine powder of crystal with uniform particle size. The measurement was carried out using KDP as reference material for the SHG conversion efficiency. The SHG efficiencies of L-argininium phosphite and KDP crystals were found to be 20 mV and 24 mV, respectively. Thus, SHG value of grown crystal was found to be 0.84 times than that of KDP.

4 Conclusion

Good quality and transparent LAPI crystals were successfully grown by slow evaporation solution growth technique. Single-crystal XRD and powder XRD data confirmed that

the titled compound exists in monoclinic system having P2₁ space group. The crystal packing of LAPI crystal displays the high propensity of N–H...O hydrogen bonds, which combines the L-argininium cations and leads to infinite supra-molecular layers running parallel to 'b'-axis. An analysis of the 2D fingerprint plot and 3D molecular Hirshfeld surface summarized all the intermolecular contacts present in the LAPI crystal system. Indeed, O...H/H...O and H...H interactions possess the most significant contribution to the overall Hirshfeld surface of both the moieties. The FWHM of the diffracted curve was 46" for the plane (200) which reveals that the compound has good crystalline perfection. The observed lower UV cutoff wavelength and bandgap for the grown crystal was nearly about 240 nm and 5.1 eV. The crystal sustainability towards the high laser beam was found to be 0.63 and 0.55 GW/cm² for single shot and multiple shots, respectively. Through SDT, no cracks were observed even after the 12th shock. Vickers microhardness analysis confirmed the softer nature of LAPI crystal. Void analysis revealed that the LAPI crystal possesses uniformly distributed void channels indicating a further scope of tailoring its anisotropic characteristics. Its piezoelectric coefficient was also measured which was found to be 12 pC/N along the (010) plane and was higher than some of the reported material which makes it suitable for energy-harvesting applications.

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Compliance with ethical standards

Conflict of interest There are no conflicts of interest to declare.

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